

Linear predictive coding

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One popular technique
of analyzing certain
physical signals

Many problems in electrical engineering require analyzing physical signals. Usually, the reason is to obtain a useful, economical model of relevant aspects of the signal. For example, we analyze radar signals to estimate the distance, size, and velocity of objects which cause the signal to be reflected back to the transmitter. Biomedical signals, such as an EEG (electro encephalo gram), contain information about the health of the person from which they were obtained. For efficient coding or storage, speech signals are often modeled using parameters of the presumed vocal tract shape generating them. In all these cases, it is important to analyze signals accurately and quickly. One popular technique for analysis of certain physical signals is linear predictive coding (LPC).

Signals produced by a relatively slowly-varying linear filtering process are most suitable for LPC, especially, if the filter is excited by infrequent, brief pulses. The future values of such signals are well estimated by a linear predictor, which outputs values based on a linear combination of previous signal values. For example, an exponentially-damped sinusoid is characterized by two parameters, its oscillation frequency and the decay time constant; ignoring excitation, a periodically sampled signal output of this second-order system is completely determined by two predictor coefficients and the signal values at two prior samples.

A Fourier transform or frequency representation can highlight important aspects of a signal. In particular, a signal's spectral magnitude, as opposed to phase, is widely used (e.g., in human perception, it is more important to model the magnitude of speech spectra accurately than their phase). LPC derives a compact yet precise representation of spectral magnitude for signals of brief duration, and has

relatively simple computation. LPC is a form of parametric (model-based) analysis which allows more accurate spectral resolution than the nonparametric Fourier transform when the signal is stationary for only a brief time. However, LPC requires imposition of a model whose type and order must correspond to the signal for best results.

For analysis of speech, LPC can help estimate the recurrence of vocal fold vibration, the shape of the vocal tract, and the frequencies and bandwidths of spectral poles and zeros (e.g., vocal tract resonances). But it primarily provides a small number of speech parameters (called LPC coefficients) that relate to the configuration of the vocal tract and thus the sound being uttered. These coefficients can be used in digital filters as multiplier values to generate a synthetic version of the original signal, or can be stored as templates for pattern recognition.

Basic principles

Assume that a signal of interest has been produced by a source exciting a linear filter. In radar, the source is a transmitted pulse waveform. So the filter can be modeled as a weighted delay line which returns the reflected and attenuated signal. In seismic analysis, an explosion provides the source, and various geophysical structures act as filters. For speech modeling, the source can represent periodic puffs of air passing through the glottis (space between the vocal cords), or a noisy waveform produced at a narrow constriction in the vocal tract and the filter corresponds to the upper vocal tract.

Consider a signal $s(t)$ sampled in time every T s (e.g., 0.0001 s for speech, or 0.01 s in seismics), so that $s(n) = s(nT)$ for integer n . It has a spectral representation similar to the Laplace transform, called the z -transform $S(z)$. The assumed production model for $s(n)$ consists of an excitation source $U(z)$ providing input to a

spectral shaping filter $H(z)$, to yield $S(z) = U(z)H(z)$. LPC analysis of $s(n)$ "deconvolves" excitation and filter estimates $\hat{U}(z)$ and $\hat{H}(z)$, following certain constraints, so that their product is as close as possible in a mean-square sense to the original $S(z)$. (We use $\hat{}$ notation to indicate estimation, e.g., $\hat{h}(n)$ is the estimated impulse response of a physical system generating $s(n)$ whose actual response $h(n)$ is not directly observable.)

To simplify the modeling problem, $\hat{U}(z)$ is often assigned a flat spectral envelope so that relevant spectral detail is confined to $\hat{H}(z)$. A flat source spectrum is a reasonable assumption for speech since the excitation for noisy "unvoiced" sounds resembles white noise. For periodic "voiced" sounds, the source is viewed as a uniform sample train, periodic in N samples (the pitch period of vocal fold vibration), having a line spectrum with uniform-area harmonics.

To simplify obtaining $\hat{H}(z)$ given a signal $s(n)$, we consider $s(n)$ to be stationary (not changing its underlying characteristics) during a "window" or "frame" of N samples (e.g., 0.01–0.03 s for speech). The $H(z)$ filter can then be modeled with constant coefficients (to be updated for each successive frame of data). $H(z)$ is assumed to have p poles and q zeros in the general case. This means that a model signal sample $\hat{s}(n)$ is a linear combination of the p previous output samples and $q + 1$ previous input samples of an LPC synthesizer:

$$\hat{s}(n) = \sum_{k=1}^p a_k \hat{s}(n-k) + G \sum_{l=0}^q b_l u(n-l)$$

where G is a gain factor for the excitation (assuming $b_0 = 1$). Equivalently,

$$\hat{H}(z) = \frac{\hat{S}(z)}{\hat{U}(z)} = G \frac{1 + \sum_{l=1}^q b_l z^{-l}}{1 - \sum_{k=1}^p a_k z^{-k}}.$$

Most LPC work assumes an all-pole model (also known as an *autoregressive*, or AR, model), where $q = 0$. An all-zero model ($p = 0$) is called a *moving average* (MA) model since the output is a weighted average of the $q + 1$ prior inputs. The more general LPC model with both poles and zeros ($p, q > 0$) is known as an autoregressive moving average (ARMA) model. AR LPC requires only solution of linear equations, while MA versions lead to nonlinear equations. Although ARMA LPC can yield a more accurate model, the simplifying all-pole assumption, i.e., that the signal spectrum is well-modeled by resonances only, is satisfactory in most applications. Physical signals tend to be characterized more by the presence of energy at certain frequencies rather than by its absence. That is, signals with spectral peaks are more common than ones with zeros. For example, the vocal tract acts as an acoustic tube of varying area which is characterized by about one resonance per kHz of bandwidth. Zeros in the speech spectrum are due to alternate paths taken, for example, into the lungs or nasal cavity rather than out of the mouth. Since hearing systems are more sensitive to spectral peaks than valleys, the effects of such zeros can be approximated by poles in an all-pole model.

If $s(n)$ is filtered by an inverse or

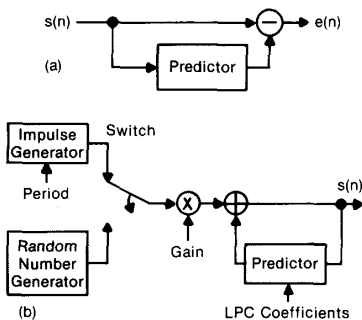


Fig. 1. All-pole LPC: (a) analysis; (b) synthesis using parametric excitation.

predictor filter (the inverse of an all-pole synthesis filter $\hat{H}(z)$),

$$A(z) = 1 - \sum_{k=1}^p a_k z^{-k},$$

the output $e(n)$ is called an *error* or *residual* signal:

$$e(n) = s(n) - \sum_{k=1}^p a_k s(n - k).$$

$E(z)$ and $\hat{U}(z)$ should be similar to the extent that $\hat{H}(z)$ is an adequate model. In EEG analysis, discontinuities in $e(n)$ can be related to brain disorders. In voiced speech, the error signal normally has sharp peaks separated in time by pitch periods (corresponding to vocal fold closures, which excite the vocal tract); this allows direct estimation of periodicity parameters. In unvoiced speech, the residual resembles white noise, and LPC synthesizers utilize pseudo-random number generators to simulate such an excitation. Since most physical signals cannot be fully modeled by a p -pole filter $\hat{H}(z)$, there will be differences between $e(n)$ and the presumed impulse train or white noise $\hat{u}(n)$.

In synthesizing a version $\hat{s}(n)$ of the original $s(n)$ using LPC coefficients, $e(n)$ can be parametrized for economy. For speech, $e(n)$ is usually represented by one bit signifying whether the speech during the frame under analysis is periodic, and, if so, by about six bits to code the period duration. The excitation for voiced speech in LPC synthesis is typically a simple train of samples spaced at the pitch period rate. The energy of these impulses (corresponding to gain factor G) is chosen to match the energy of the residual signal (Fig. 1). A set of 10 LPC coefficients requires about 50 bits when quantized individually, although coding them as a group (vector quantization) can reduce that to about 10 bits. At a typical frame rate of 50/s, intelligible speech signals can be reconstructed from less than 1 kbit/s. The usual data rate for logarithmic PCM (pulse code modulation), as used in the telephone network, is 64 kbits/s, but the speech is of higher "toll quality" (LPC's quality is synthetic but intelligible, with a buzzy mechanical sound). Differential PCM coders, especially ones which adapt the predictor dynamically to reflect signal changes (ADPCM), offer a compromise between LPC and log PCM. ADPCM can yield toll-quality by replacing the low-bit-rate parametric excitation of Fig. 1b with a coded version of $e(n)$. LPC and ADPCM calculate the same spectral coefficients, but LPC also parametrizes $e(n)$ while ADPCM simply codes it (with PCM). To bridge the gap between about 2 kbits/s for basic LPC and 24 kbits/s for ADPCM, a new method called *multipulse* LPC models $e(n)$

with about 8 pulses/pitch period, chosen for their perceptual relevance.

Least-squares methods

Two approaches can obtain the LPC coefficients a_k characterizing an all-pole $\hat{H}(z)$ model. The classical *least-squares* method selects a_k to minimize the mean energy in $e(n)$ over a frame of signal data, while the *lattice* approach permits instantaneous updating of the coefficients. In the former technique, either $s(n)$ or $e(n)$ is windowed to limit the extent of the signal under analysis. Repeated analysis of successive frames of data permits following spectral changes in a time-varying signal.

The first of two common least-squares techniques is the *autocorrelation* method, which multiplies the signal by a Hamming or similar time window $w(n)$ so that $x(n) = w(n)s(n)$ has finite duration (N samples). Thus, $x(n) = 0$ outside the range $0 \leq n \leq N - 1$. LPC models all $x(n)$ samples within each frame; thus when the signal is nonstationary, the LPC coefficients describe a smoothed average.

Let E be the energy in the error:

$$E = \sum_{n=-\infty}^{\infty} e^2(n) = \sum_{n=-\infty}^{\infty} \left[x(n) - \sum_{k=1}^p a_k x(n - k) \right]^2$$

where $e(n)$ is the residual corresponding to the windowed signal $x(n)$. The values of a_k that minimize E are found by setting $\partial E / \partial a_k = 0$ for $k = 1, 2, 3, \dots, p$. This yields p linear equations ($i = 1, 2, \dots, p$)

$$\sum_{n=-\infty}^{\infty} x(n - i) x(n) = \sum_{k=1}^p a_k \sum_{n=-\infty}^{\infty} x(n - i) \cdot x(n - k)$$

in p unknowns a_k . Since the first term as the autocorrelation $R(i)$ of $x(n)$ and $x(n)$ has finite duration,

$$\sum_{k=1}^p a_k R(i - k) = R(i), \quad 1 \leq i \leq p,$$

$$\text{where } R(i) = \sum_{n=i}^{N-1} x(n) x(n - i).$$

Another least-squares technique called the *covariance* method windows the error $e(n)$ instead of $s(n)$:

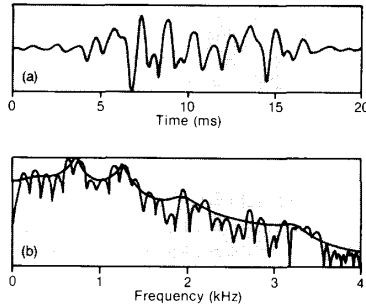


Fig. 2. (a) Speech signal (Hamming windowed); (b) spectral magnitude via Fourier transform (ragged line) and via 12-pole autocorrelation LPC (smooth line).

$$E = \sum_{n=-\infty}^{\infty} e^2(n) w(n).$$

Usually the error is weighted uniformly in time via a simple rectangular window of N samples, which in effect replaces $R(i)$ above with the covariance function

$$\phi(i, k) = \sum_{n=0}^{N-1} s(n-k) s(n-i).$$

The two techniques vary in windowing effects, which lead to differences in computation and stability of synthesis filters. The autocorrelation approach introduces distortion into the spectral estimation procedure since time windowing rolls together the original short-time spectrum with the frequency response of the window. Most windows have lowpass frequency responses; thus, the spectrum of the windowed signal is a smoothed version of the original. The extent and type of smoothing depends on the window shape and duration.

Computational factors

In the autocorrelation method, the p linear equations to be solved can be viewed in matrix form as $\mathbf{R}\mathbf{a} = \mathbf{r}$, where $\mathbf{R}$ is a $p \times p$ matrix of elements $R(i, k) = R(|i - k|)$, ($1 \leq i, k \leq p$), $\mathbf{r}$ is a column vector ($R(1)$, $R(2)$, $\dots$, $R(p)$)^T, and $\mathbf{a}$ is a column vector of LPC coefficients (a_1 , a_2 , $\dots$, a_p)^T. Computing the LPC vector directly requires inversion of the $\mathbf{R}$ matrix and multiplication of the resultant $p \times p$ matrix with the $\mathbf{r}$ vector. A parallel situation occurs for the covariance approach.

Redundancies in the $\mathbf{R}$ and Φ matrices allow efficient calculation of the LPC coefficients without explicitly inverting a $p \times p$ matrix. In particular, both matrices are symmetric (e.g., $\phi(i, k) = \phi(k, i)$). However, $\mathbf{R}$ is also Toeplitz (all elements along a given

diagonal are equal), where Φ is not. As a result, the autocorrelation approach is simpler computationally than the common covariance method. One must emphasize, however, that if $N \gg p$ (often the case in speech processing), then computation of the $\mathbf{R}$ or Φ matrix dominates overall. The frame size N should be large enough to average out undesired components (e.g., background noise) and allow good spectral resolution. Yet, the size should be small enough to minimize computation and avoid smearing out relevant transients in the input signal. (For speech, N often exceeds 100, while p is about 10.)

The additional redundancy in the Toeplitz $\mathbf{R}$ matrix allows the efficient Levinson-Durbin recursive procedure, in which the following set of ordered equations is solved recursively for $m = 1, 2, \dots, p$:

$$k_m = \frac{R(m) - \sum_{i=1}^{m-1} a_{m-1}(i) R(m-i)}{E_{m-1}},$$

$$a_m(m) = k_m,$$

$$a_m(i) = a_{m-1}(i) - k_m a_{m-1}(m-i),$$

$$\text{for } 1 \leq i < m,$$

$$E_m = (1 - k_m^2) E_{m-1},$$

where initially $E_0 = R(0)$ and $a_0 = 0$. At each cycle m , the coefficients $a_m(i)$ (for $i = 1, 2, \dots, m$) describe the optimal m th-order linear predictor, and the minimum error E_m is reduced by the factor $(1 - k_m^2)$. (This is useful when the optimal order is not known *a priori*.) Since E_m , a squared error, is never negative, $|k_m| \leq 1$. This condition on the reflection coefficients k_m also guarantees a stable LPC synthesis filter $\hat{H}(z)$ since no roots of $A(z)$ are outside the unit circle in the z -plane. Unlike the covariance method, the autocorrelation method guarantees a stable synthesis filter. The finite wordlengths of digital computers may result in some k_m with magnitude exceeding unity. However, stability is easily obtained by reducing $|k_m|$ in these cases. Such a simple stability test is not available for the a_k coefficients, although cases of instability can be reduced by increasing N .

Instantaneous updating

In least-squares LPC modeling, $s(n)$ is divided into successive frames of data and spectral coefficients are obtained for each frame. Alternatively, the LPC parameters can be determined sample by sample, updating the model for each signal sample. This is

attractive for real-time implementations (e.g., echo cancellation in the telephone network) because it reduces the delay inherent in the frame approach. In instantaneous LPC estimation, a recursive procedure is necessary to minimize computation. Each new sample updates some intermediate signal measures (e.g., a local energy or covariance measure), from which the LPC parameters are revised. Recalculating and inverting the covariance matrix for each signal sample is unnecessary.

Two basic ways to implement a linear predictor are the *transversal* form (i.e., direct-form digital filter, using the a_k coefficients as multiplier values) and the *lattice* form (Fig. 3). The transversal predictor directly updates N LPC coefficients $a_k(n)$ (the k th spectral coefficient at time n) as follows:

$$a_k(n+1) = v a_k(n) + (1-v) a_k^* + G_k(n+1) e(n+1),$$

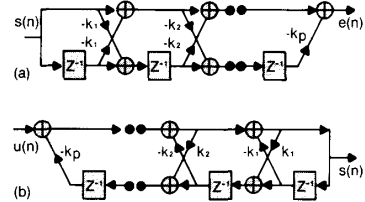


Fig. 3. Lattice filters: (a) inverse filter $A(z)$, (b) synthesis filter $1/A(z)$.

where $\mathbf{a}^*$ is a target vector of coefficients that is approached exponentially (depending on the damping factor v) in silence (i.e., when the LPC error $e(n) = 0$) and $\mathbf{G}$ is an "automatic gain control" vector (based on the N previous signal samples) that controls the model adaptation. The *gradient* or *least-mean-square* (LMS) approach assigns simple values to $\mathbf{G}$:

$$G_k(n) = \frac{\hat{s}(n-k)}{C + \sum_{i=0}^{N-1} w^i \hat{s}^2(n-i-1)},$$

where the denominator is simply a recent signal energy estimate (with weighting controlled by a damping factor w) and C is a constant to avoid division by zero during a null signal. Alternatives to the gradient approach, e.g., the *Kalman algorithm*, yield a more precise spectral model but need more computation for $\mathbf{G}$.

Spectral estimation

Parseval's theorem for a discrete-time signal (e.g., the error signal $e(n)$)

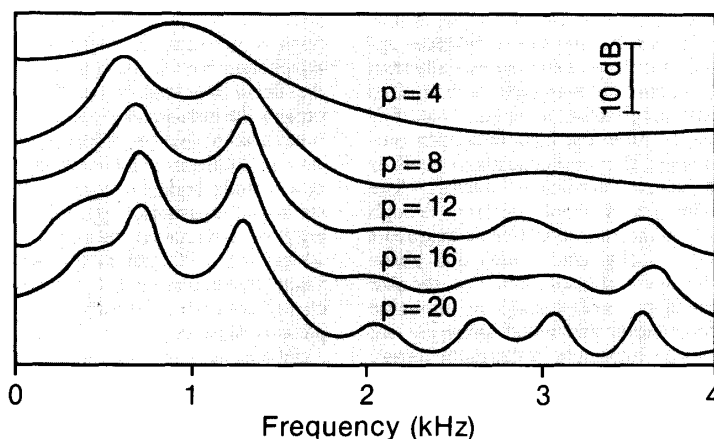


Fig. 4. LPC spectra for speech of Fig. 2 using 4, 8, 12, 16, and 20 poles.

and its Fourier transform is

$$E = \sum_{n=-\infty}^{\infty} e^2(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} |E(e^{j\omega})|^2 d\omega. \quad (1)$$

Since $e(n)$ can be obtained by passing $s(n)$ through the inverse LPC filter $A(z) = G/H(z)$, the residual error can be expressed as

$$E = \frac{G^2}{2\pi} \int_{-\pi}^{\pi} \frac{|S(e^{j\omega})|^2}{|\hat{H}(e^{j\omega})|^2} d\omega. \quad (2)$$

Obtaining the LPC coefficients by minimizing E is equivalent to minimizing the average ratio of the signal spectrum to its LPC approximation. Equal weight is given to all frequencies, but $|\hat{H}(e^{j\omega})|$ models the peaks in $|S(e^{j\omega})|$ better than the valleys (Fig. 2b) because the contribution to the error E at frequencies where the signal spectrum exceeds its LPC approximation is greater than for the opposite condition. $|\hat{H}(e^{j\omega})|$ tends to follow the spectral envelope of $|S(e^{j\omega})|$ just below the harmonic peaks. Because LPC emphasizes peaks over valleys (which also reflects the importance of peaks in most physical signals), the bandwidths of signal resonances are less accurately represented than their center frequencies. This is not a major problem in coding speech since the just noticeable difference for bandwidths is about ten times that for resonance frequencies.

The spectrum $|\hat{H}(e^{j\omega})|$ of an all-pole LPC model is limited, by the number of poles p used, in the degree of spectral detail it can model in $|S(e^{j\omega})|$. To model sine waves or resonances (e.g., in a background of noise), LPC should use at least two poles per resonance. Rapid spectral

variations (e.g., due to noise, or too many harmonics in the case of a signal where the main interest lies in its smooth spectral envelope) cannot be completely modeled by a low-order $|\hat{H}(e^{j\omega})|$.

The choice of the order p for the LPC model is a compromise among spectral accuracy, computation time and memory, and transmission bandwidth (the last being relevant only for coding or storage applications). In the limit as $p \rightarrow \infty$, $|\hat{H}(e^{j\omega})|$ matches $|S(e^{j\omega})|$ exactly (Fig. 4), but at the cost of increased memory and computation. In general, there should be a sufficient number of poles to represent all relevant resonances in the signal bandwidth plus a few additional poles to approximate possible zeros in the spectrum as well as general spectral shaping (e.g., 10 poles for 8-kHz sampled speech). Too high an order can lead to spurious spectral peaks of little use in characterizing the signal. Non-speech applications often use low-order LPC (e.g., $p = 1 - 4$); in radar clutter analysis, the spatial characteristics of weather or ground objects are suitably modeled with a few poles, for the purpose of identification or signal enhancement.

Finally, note that the choice of mean square error as a distortion measure leads to modeling the spectral magnitude of a signal and not its phase. The loss of phase information in LPC modeling, as exemplified in the use of autocorrelation data (which eliminates phase) to represent a signal, is acceptable in most applications. For example, in speech recognition, the LPC coefficients capture the vocal tract shape, which is important to distinguish different sounds; the discarded phase primarily represents timing information in the glottal excitation,

which is of little use in discriminating phonemes. Phase can be important, however, for resynthesized speech from LPC parameters; LPC speech sounds synthetic, while ADPCM speech (which preserves phase by coding the error signal) does not.

Integrated circuits

LPC analysis is complex enough to require digital signal processing chips, e.g., the NEC 7720 or the TI TMS320, which do single-cycle multiplies (unlike standard microprocessors). LPC synthesis, on the other hand, involves only a digital filter of typically 10 stages (each stage using 1–2 multiplies at 8000–10,000 samples/s) plus excitation generation. Pipelined multiplies taking a few microseconds suffice, and both the program and memory code are small. For speech, besides the period and gain, only about 10 reflection coefficients and their corresponding signal values need to be stored in data RAM (random access memory). Inexpensive LSI chips for LPC synthesis have up to 32 kbits of internal ROM or can access up to 128 kbits of external ROM. The lattice filter method is prevalent because of its guaranteed stability and suitability for fixed-point arithmetic. Most chips use a simple impulse train for periodic excitation, but others store an oversampled waveform that is adjusted for each period duration.

Read more about it

- A good reference on LPC is "Linear prediction: a tutorial review," J. Makhoul (1975) *IEEE Proc.*, vol. 63, pp. 561–580, 1975. See also articles by M. Schroeder and J. Gibson, *IEEE ASSP Magazine*, vol. 1, no. 3, 1984.
- An article by S. Kay and S. Marple puts LPC in the broader view of spectral estimation. "Spectrum analysis—a modern perspective," *IEEE Proc.*, vol. 69, pp. 1380–1418, 1981.
- For an introduction to issues of speech processing, see D. O'Shaughnessy, *Speech Communication*. Reading, MA: Addison-Wesley, 1987.

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